Integration: Inverse Hyperbolic Functions

Questions 1–57

Question 1. Use the substitution $x = \sinh u$ to show that

$$\int \frac{1}{\sqrt{x^2+1}}\,dx = \operatorname{arsinh} x + c.$$

Question 2. Use the substitution $x = \cosh u$ to show that

$$\int \frac{1}{\sqrt{x^2-1}}\,dx = \operatorname{arcosh} x + c.$$

Question 3. Use the substitution $x = \tanh u$ to show that

$$\int \frac{1}{1-x^2}\,dx = \operatorname{artanh} x + c.$$

Question 4. Find:

a) $\displaystyle\int \frac{4}{\sqrt{x^2-1}}\,dx$

b) $\displaystyle\int \frac{1}{2\sqrt{x^2+1}}\,dx$

c) $\displaystyle\int \frac{6}{1-x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{4x^2-1}}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9x^2+1}}\,dx$

f) $\displaystyle\int \frac{1}{1-25x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{16x^2-1}}\,dx$

h) $\displaystyle\int \frac{\sqrt{8}}{\sqrt{2x^2+1}}\,dx$

i) $\displaystyle\int \frac{100}{1-\frac{1}{5}x^2}\,dx$

Question 5.

a) Show that

$$\frac{1}{15-x^2} \equiv \frac{1}{15(1-(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{15-x^2}\,dx$.

Question 6.

a) Show that

$$\frac{1}{\sqrt{x^2+3}} \equiv \frac{1}{\sqrt{3}\sqrt{(kx)^2+1}},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{\sqrt{x^2+3}}\,dx$.

Question 7. Find:

a) $\displaystyle\int \frac{1}{25-x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2+9}}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{x^2-4}}\,dx$

d) $\displaystyle\int \frac{1}{8-x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{x^2+12}}\,dx$

f) $\displaystyle\int \frac{1}{\sqrt{x^2-20}}\,dx$

g) $\displaystyle\int \frac{5}{10-x^2}\,dx$

h) $\displaystyle\int \frac{14}{\sqrt{x^2+7}}\,dx$

i) $\displaystyle\int \frac{18}{\sqrt{x^2-3}}\,dx$

j) $\displaystyle\int \frac{1}{9-4x^2}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{9x^2+4}}\,dx$

l) $\displaystyle\int \frac{1}{\sqrt{25x^2-16}}\,dx$

m) $\displaystyle\int \frac{15}{4-\frac{1}{4}x^2}\,dx$

n) $\displaystyle\int \frac{3}{\sqrt{16x^2+\frac{1}{9}}}\,dx$

o) $\displaystyle\int \frac{18}{\ldots}$

Question 8. Given that a and k are constants, find:

a) $\displaystyle\int \frac{1}{a^2-x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2+a^2}}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{k^2x^2-a^2}}\,dx$

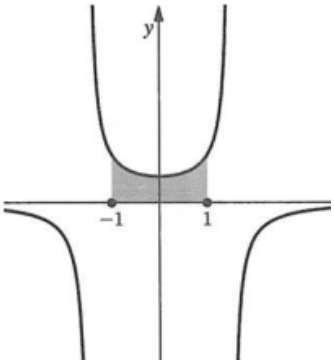
Question 9. Find the exact values of:

a) $\displaystyle\int_{-1/2}^{1/2} \frac{1}{1-x^2}\,dx$

b) $\displaystyle\int_{1}^{2} \frac{1}{\sqrt{4x^2-1}}\,dx$

c) $\displaystyle\int_{-5}^{5} \frac{30}{\sqrt{36x^2+25}}\,dx$

Question 10. The diagram below shows the curve $y = \frac{3}{9-4x^2}$. Show that the exact shaded area is $\frac{1}{2}\ln 5$.



Question 11. Show that

$$\int_{3/5}^{4/5} \left(\frac{5}{\sqrt{25x^2-4}} + \frac{31}{155x-92} \right) dx = \ln(6 - 2\sqrt{5} + 3\sqrt{3} - \sqrt{15}).$$

Question 12. Show that

$$\int_{1}^{3/2} \frac{4}{\sqrt{4x^2-1}}\, dx = \ln(a + b\sqrt{6} + c\sqrt{3} + d\sqrt{2}),$$

where a, b, c and d are integers to be found.

Question 13. It is given, for $x \in \mathbb{R}$, that

$$f(x) = \frac{19 - 39x^2}{(1 - x^2)(4 - 9x^2)}.$$

a) Write $f(x)$ in the form $\dfrac{Ax + B}{1 - x^2} + \dfrac{Cx + D}{4 - 9x^2}$, where A, B, C and D are constants to be found.

b) Show that $\displaystyle\int f(x)\,dx = p\,\text{artanh}(qx) + r\,\text{artanh}(sx) + c$, where p, q, r and s are constants to be found, and c is a constant of integration.

Question 14. Find:

a) $\displaystyle\int \frac{12x^2 + 1}{(1 - 9x^2)(1 - 16x^2)}\,dx$

b) $\displaystyle\int \frac{10x^2 + 47}{(1 - 4x^2)(25 - x^2)}\,dx$

c) $\displaystyle\int \frac{108x^4 + 12x^3 - 237x^2 - 27x + 25}{(1 - 9x^2)(9 - 4x^2)}\,dx$

Question 15. Given that

$$\frac{1-2x}{1-x^2} \equiv \frac{1}{1-x^2} - \frac{2x}{1-x^2},$$

find $\displaystyle\int \frac{1-2x}{1-x^2}\,dx$.

Question 16. Find:

a) $\displaystyle\int \frac{6x+1}{1-9x^2}\,dx$

b) $\displaystyle\int \frac{2-8x}{4-x^2}\,dx$

c) $\displaystyle\int \frac{8-3x}{16-9x^2}\,dx$

d) $\displaystyle\int \frac{10x-4}{25-100x^2}\,dx$

Question 17. Given

$$\frac{1+x}{\sqrt{x^2-1}} \equiv \frac{1}{\sqrt{x^2-1}} + \frac{x}{\sqrt{x^2-1}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{x^2-1}}\,dx$.

Question 18. Given

$$\frac{4x+1}{\sqrt{x^2+1}} \equiv \frac{4x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}},$$

find $\displaystyle\int \frac{4x+1}{\sqrt{x^2+1}}\,dx$.

Question 19. Find:

a) $\int \frac{2x-1}{\sqrt{4x^2-1}}\,dx$

b) $\int \frac{1-3x}{\sqrt{9x^2+1}}\,dx$

c) $\int \frac{4+6x}{\sqrt{9x^2-4}}\,dx$

d) $\int \frac{8x-3}{\sqrt{4x^2+36}}\,dx$

Question 20. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \frac{1}{\sqrt{x^2+1}}\,dx$

b) $\int \frac{1}{\sqrt{x^2-1}}\,dx$

c) $\int \frac{1}{1-x^2}\,dx$

d) $\int \frac{1}{\sqrt{4x^2+1}}\,dx$

e) $\int \frac{1}{\sqrt{x^2-9}}\,dx$

f) $\int \frac{1}{1-9x^2}\,dx$

g) $\int \frac{1}{\sqrt{x^2+25}}\,dx$

h) $\int \frac{1}{\sqrt{36x^2-1}}\,dx$

i) $\int \frac{1}{100-x^2}\,dx$

j) $\int \frac{1}{\sqrt{49x^2+1}}\,dx$

k) $\int \frac{1}{\sqrt{x^2-64}}\,dx$

l) $\int \frac{1}{1-144x^2}\,dx$

m) $\int \frac{1}{\sqrt{4x^2+9}}\,dx$

n) $\int \frac{1}{\sqrt{9x^2-16}}\,dx$

o) $\int \frac{1}{4-25x^2}\,dx$

p) $\int \frac{1}{\sqrt{36x^2+25}}\,dx$

Question 21. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 + 1)^{3/2}\, dx$

Question 22. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\int \sqrt{x^2 - 1}\, dx$

b) $\int (x^2 - 1)^{3/2}\, dx$

Question 23. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1-x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(1-9x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1-x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(4-x^2)^3}\,dx$

Question 24. For each of the following integrals, suggest a suitable hyperbolic substitution and then integrate:

a) $\displaystyle\int \sqrt{25x^2+1}\,dx$

b) $\displaystyle\int (x^2+9)^{3/2}\,dx$

c) $\displaystyle\int \sqrt{x^2-36}\,dx$

d) $\displaystyle\int (81x^2-1)^{3/2}\,dx$

Question 25. Complete the square for each of these:

a) $x^2 + 4x + 3$
b) $x^2 - \frac{10}{3}x + \frac{34}{9}$
c) $4x^2 + 24x + 37$
d) $9x^2 - 18x + 8$
e) $x^2 + 6x + 18$
f) $9x^2 - 36x + 32$

Question 26.

a) Write $x^2 + 20x + 101$ in the form $(x + p)^2 + q$, where p and q are integers.
b) Hence, using the substitution $x + 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 + 20x + 101}}\, dx$.

Question 27.

a) Write $x^2 - 14x + 48$ in the form $(x + p)^2 + q$, where p and q are integers.

b) Hence, using the substitution $x - 7 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{x^2 - 14x + 48}}\, dx$.

Question 28. Find:

a) $\displaystyle\int \frac{1}{\sqrt{x^2 - 10x + 26}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{x^2 + \frac{2}{3}x - \frac{8}{9}}}\, dx$

Question 29. Find the exact mean value of $f(x)$ over the interval $\left[-\frac{2}{5}, \frac{2}{5}\right]$, given

$$f(x) = \frac{1}{\sqrt{x^2 - \frac{4}{5}x + \frac{29}{25}}}, \qquad x \in \mathbb{R}.$$

Question 30.

a) Write $4x^2 - 40x + 101$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $2x - 10 = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 40x + 101}}\, dx$.

Question 31.

a) Write $16x^2 + 96x + 143$ in the form $p(x + q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $4x + 12 = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{16x^2 + 96x + 143}}\, dx$.

Question 32. Find:

a) $\displaystyle\int \frac{1}{\sqrt{9x^2 - 144x + 575}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{2x^2 + 12x + 19}}\, dx$

Question 33.

a) Write $4x^2 - 56x + 205$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}(x-7) = \sinh u$, find $\displaystyle\int \frac{1}{\sqrt{4x^2 - 56x + 205}}\, dx$.

Question 34.

a) Write $3x^2 + 6x - 1$ in the form $p(x+q)^2 + r$, where p, q and r are integers.

b) Hence, using the substitution $\frac{\sqrt{3}}{2}(x+1) = \cosh u$, find $\displaystyle\int \frac{1}{\sqrt{3x^2 + 6x - 1}}\, dx$.

Question 35. Find:

a) $\displaystyle\int \frac{1}{\sqrt{25x^2 - 100x + 91}}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{5x^2 + 10\sqrt{5}x + 29}}\,dx$

Question 36. Show that

$$\int_{-1}^{1} \frac{1}{\sqrt{4x^2 + 8x + 13}}\,dx = \frac{1}{2}\ln k,$$

where k is a positive integer to be found.

Question 37. Complete the square for each of these:

a) $-3 - 4x - x^2$

b) $-24 + 10x - x^2$

c) $-15 - 16x - 4x^2$

d) $-143 + 72x - 9x^2$

e) $-45 - 14x - x^2$

f) $8 + 4x - 4x^2$

Question 38.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$, where p and q are integers.

b) Hence, using the substitution $x - 8 = \tanh u$, find $\displaystyle\int \frac{1}{(-63 + 16x - x^2)}\, dx$.

Question 39. Find:

a) $\displaystyle\int \frac{1}{(-99-20x-x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-168+26x-x^2)}\,dx$

Question 40.

a) Write $-35+24x-4x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $2x-6=\tanh u$, find $\displaystyle\int \frac{1}{(-35+24x-4x^2)}\,dx$.

Question 41. Find:

a) $\displaystyle\int \frac{1}{(-224-90x-9x^2)}\,dx$

b) $\displaystyle\int \frac{1}{(-24+50x-25x^2)}\,dx$

Question 42.

a) Write $11+10x-x^2$ in the form $p-(x+q)^2$, where p and q are integers.

b) Hence, using the substitution $\frac{x-5}{6}=\tanh u$, find $\displaystyle\int \frac{1}{(11+10x-x^2)}\,dx$.

Question 43. Find:

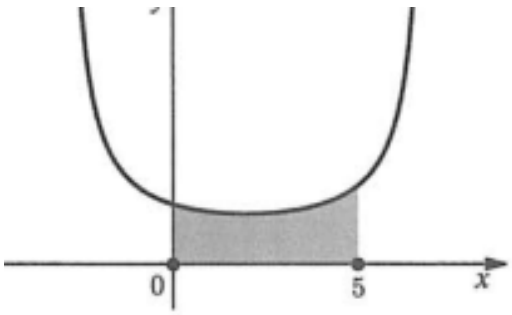
a) $\int \frac{1}{(91-18x-9x^2)}\,dx$

b) $\int \frac{1}{(-20-24x-4x^2)}\,dx$

Question 44. The curve shown in the diagram below has equation

$$y = \frac{5}{63+12x-3x^2}.$$

Find the exact area of the shaded region.



Question 45. A curve has equation

$$y = \frac{2}{\sqrt{-16 + 16x - 3x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, and the lines $x = 2$ and $x = 3$ is rotated through 2π radians about the x-axis.

Question 46. Use the substitution $x = \cosh u$ to find $\displaystyle\int \sqrt{x^2 - 1}\, dx$.

Question 47. Use the substitution $x = \sinh u$ to find $\int \sqrt{x^2 + 1}\, dx$.

Question 48. Use a suitable substitution to find $\int \sqrt{9x^2 - 1}\, dx$.

Question 49. Find the exact value of

$$\int_{12}^{13} \sqrt{x^2 - 144}\, dx.$$

Question 50. Show that

$$\int_0^2 \sqrt{25x^2 + 1}\, dx = \sqrt{101} + \frac{1}{10} \ln(10 + \sqrt{101}).$$

Question 51. Find:

a) $\displaystyle\int \sqrt{x^2 - 10x + 26}\, dx$

b) $\displaystyle\int \sqrt{x^2 + 6x + 8}\, dx$

Question 52. Show that

$$\int_3^5 \sqrt{x^2 - 8x + 17}\, dx = \sqrt{2} + \ln(1 + \sqrt{2}).$$

Question 53. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2-1}}\,dx$.

Question 54. Use a suitable substitution to find $\int \frac{x^2}{\sqrt{x^2+1}}\,dx$.

Question 55. Use a suitable substitution to find $\displaystyle\int \frac{x^3}{\sqrt{x^2+1}}\,dx$.

Question 56. Use the substitution $x = \sinh^2 u$ to find $\displaystyle\int \sqrt{\frac{x}{x+1}}\,dx$.

Question 57. Use the substitution $x = \cosh u$ to find $\int (x^2 - 1)^{3/2}\, dx$.